From Classroom/Integrated School Programs 7 in Top 20, 23 in Top 100, 54 in Top 300, 106 in Top 500 All India Ranks & 2314 Students from Classroom /Integrated School Programs & 3723 Students from All Programs have been Awarded a Rank in JEE (Advanced), 2013

FIITJEE

JEE(Advanced)-2014

ALL INDIA TEST SERIES

ANSWERS, HINTS & SOLUTIONS

CRT –II
(Paper-1)

Q. No.	PHYSICS	CHEMISTRY	MATHEMATICS
1.	C	C	B
2.	A	D	A
3.	C	D	B
4.	D	A	C
5.	B	D	A
6.	D	C	D
7.	C	B	D
8.	B	C	C
9.	C	C	A
10.	A	C	A
11.	B, D	D	A, B, D
12.	A, B, C	A, B, C, D	A, B, C, D
13.	A, B, C	B	A, B
14.	B, D	A, B, C, D	A, B, C
15.	A, B, D	B, C, D	A, D
1.	1	2	2
2.	7	6	3
3.	4	1	4
4.	3	2	0
5.	1	4	6

Physics

PART – I

3. $F = v_e \frac{dm}{dt} = \rho A v^2$

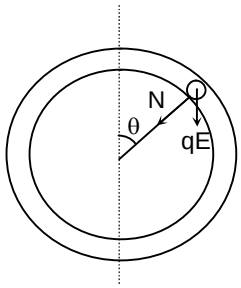
$\tau = \frac{(\rho A v^2)H}{2}$

7. $N_H + qE\cos\theta = \frac{mv^2}{R}$ …(i)

$N_H = 0$

($N_H \rightarrow$ horizontal component of normal force)

$qER(1-\cos\theta) = \frac{1}{2}mv^2$



8. $n = \frac{\left(\frac{F}{A}\right)}{\left(\frac{dv}{dx}\right)} = \frac{\frac{0.02}{10}}{\frac{2}{1}} = 10^{-3}\,kg/ms = 10^{-2}\,poise$

9. Velocity of bike V_b = 30 m/ s

Let velocity of car be V_c

$f_{obs} = f_{actual} \times \frac{V_{so} + V_b}{V_{so} - V_c}$

$120\ Hz = 100\ Hz \times \frac{330+30}{330-V_c}$

$\Rightarrow$ V_C = 30 m/s

Distance between initial and final pulse of horn

= $V_{so} \times \Delta t - V_C\, \Delta t$ {Δt is duration of emission of sound}

= (330 – 30) × 6 = 1800 m

This wave train travels with 330 m/s and cross bike rider moving with velocity of 30 m/s towards it.

time taken = $\frac{length\ of\ wave\ train}{V_{so} + V_b}$ = $\frac{1800}{330+30} = \frac{180}{36} = 5$ sec

11. Kinetic friction acts opposite to relative velocity w.r.t. contact surface.

12. $X_L = 2X_C$

$X'_C = 4X_C$

$X'_L = 2X_L$

$\Rightarrow X'_L - X'_C = 2X_L - 4X_C = 2[X_L - 2X_C] = 0$

14. Since vertical Impulse is acting on A

Hence momentum of A & B will constant in horizontal direction only.

and M.E. of A & B will remain constant

SECTION –C

1. Angular momentum Conservation

$$\frac{1}{2}MR^2\omega_0 = \left(\frac{MR^2}{2} + mR^2\right)\omega'$$

$$\left(\frac{M}{M+2m}\right)\omega_0 = \omega'$$

$$U = \omega' R = \left(\frac{M}{M+2m}\right)\omega_0 R$$

2. M.E. Conservation

$$mg(4) - \frac{k}{2}x^2 = \frac{1}{2}mv^2$$

$kx = 55$

$-mg + kx = ma$

$55-20 = 2a$

$a = 17.5\ m/sec^2$

3. $\mu = \left(\frac{qBR}{mg}\right)t = 0.4$

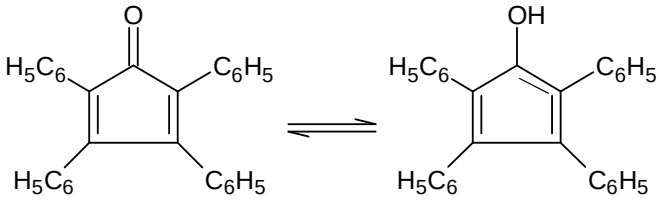
5. $t = \frac{dE[B^2\ell^2C + M]}{B\ell mg} = 1000\,sec = 1$

Chemistry

PART – II

SECTION – A

1. In (C), the tautomeric form will be



The tautomeric form, contains sp-hybridised carbon in ring, making it unstable.
$\therefore$ tautomerism is not shown by (C).

2. $2AgBr + 2OH^- + HO\text{-}C_6H_4\text{-}OH \longrightarrow 2Ag(s) + 2H_2O(\ell) + O{=}C_6H_4{=}O + 2Br^-(aq)$

Hydroquinol — Hydroquinone

3. PV^{γ} = constant

$$P_1 \times \left(\frac{105V}{100}\right)^{\gamma} = \text{Constant}$$

$$\therefore \frac{P_1}{P} \times \left(\frac{105}{100}\right)^{\gamma} = 1$$

$$\text{or } \frac{P_1}{P} = \left[\frac{100}{105}\right]^{\frac{7}{5}}$$

(γ for N_2 or O_2 = 7/5)
$P_1 = 0.93P$,
$\therefore$ 7% decrease in pressure.

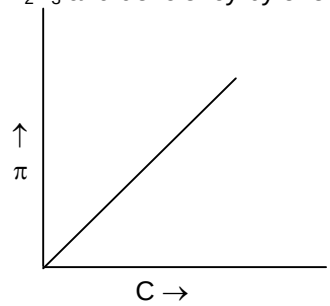
4. O^{2-} is replaced by $X^{-8/3}$, so formula of spinel is $MgAl_2X_3$ and deficiency by one anion.

5. $$\pi = \frac{W}{M} \times \frac{RT}{V}$$

$$\text{Slope} = \frac{RT}{M}$$

$$\Rightarrow M = \frac{0.082 \times 293}{4.65 \times 10^{-3} \times 10^{-3}}$$

$$\Rightarrow 5.16 \times 10^6 \text{ g}$$

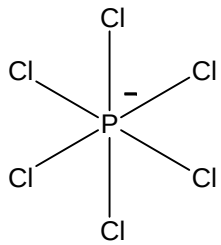


6. $PBr_5(s) \longrightarrow [PBr_4]^+ + Br^-$

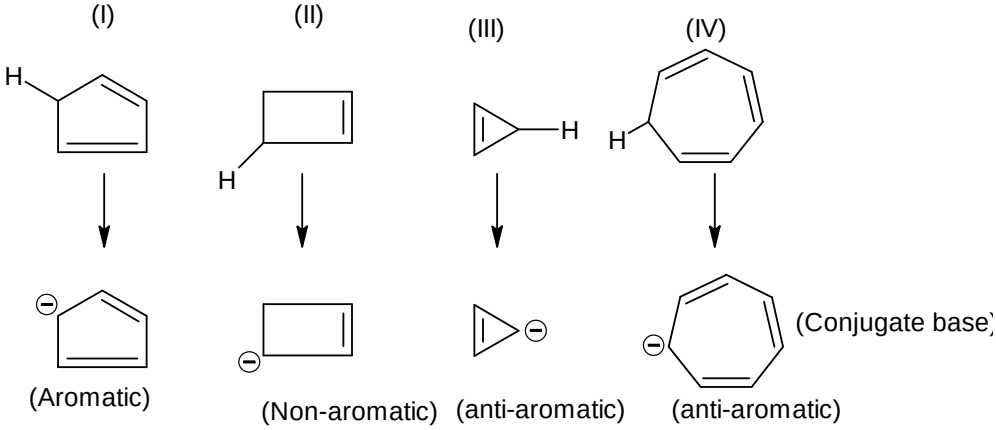
$N_2O_5(s) \longrightarrow NO_2^+ + NO_3^-\ (sp^2)$

$PCl_5(s) \longrightarrow PCl_4^+ + PCl_6^-\ (sp^3d^2)$

$Cl_2O_6(s) \longrightarrow ClO_2^+ + ClO_4^-\ (sp^3)$



7. (I) (II) (III) (IV)



(Aromatic) (Non-aromatic) (anti-aromatic) (anti-aromatic)

(III) and (IV) both are anti-aromatic, but (IV) has more resonating structures than (III).
So, stability of conjugate bases: (I) > (II) > (IV) > (III)
And pk_a order is: (I) < (II) < (IV) < (III)

8. pH = 9.71, $\therefore \left[H^+\right] = 1.95 \times 10^{-10}$

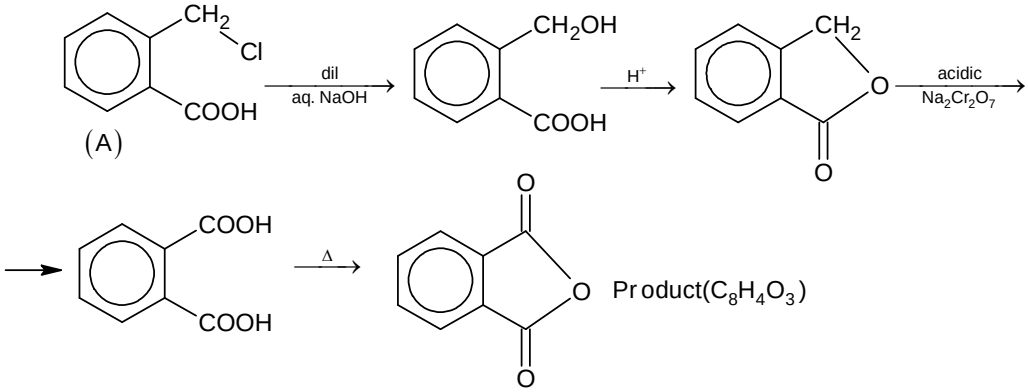
$$PrNH_2 + HCl \rightleftharpoons PrNH_3^+ + Cl^-$$

$$K_a = \frac{Pr\left[NH_2\right]\left[H^+\right]}{\left[PrNH_3^+\right]}$$

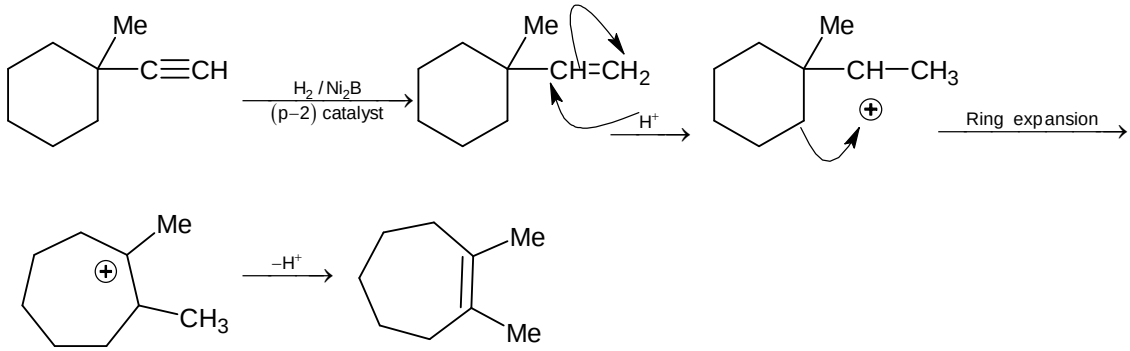
$$\frac{K_a}{\left[H^+\right]} = \frac{Pr\left[NH_2\right]}{\left[PrNH_3^+\right]}$$

$$\Rightarrow \frac{\left[PrNH_2\right]}{\left[PrNH_3^+\right]} = 0.1$$

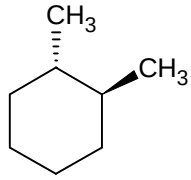
9.



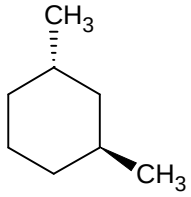
10.



11.



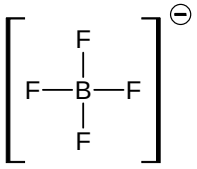
Trans - 1, 2 - dimethyl cyclohexane



Trans - 1, 3 - dimethyl cyclohexane

No plane of symmetry in both the above isomers, hence optically active.

12. $NO^+[BF_4]^-$



No. of σ bonds in $[BF_4]^- = 4$

= B. O. of $NO^+ = 3.0$, i.e. one sigma bond and two π bonds.

$\therefore$ It has 5 sigma and two ,π bonds.

$\Rightarrow NO^+$ is diamagnetic and BF_4^- is also diamagnetic

$\Rightarrow$ B—F bond energy is lower in BF_4^- than in BF_3, due to presence of back bonding in BF_3.

13. $$E_{2H^+/H_2} = E^o_{2H^+/H_2} - \frac{0.059}{2}\log_{10}\frac{p(H_2)}{[H^+]^2}$$

14. $\Delta H^o = \Delta H_f^o(\text{products}) - \Delta H_f^o(\text{reactants})$

= –166 – (–51) = – 115 KJ mol^{-1}

ΔS^o = 266 – 243 = 23J = 0.023 $KJmol^{-1}K^{-1}$

ΔH^o = –ve

ΔS^o = +ve

$\therefore \Delta G^o = \Delta H^o - T\Delta S^o$

ΔG^o will be –ve at all temperature and $\therefore$ the reaction is favour at temperature.

15. Formula of complex is : $[Co(SCN)_2(NH_3)_4]_3[Co(ox)_3]$

$\Rightarrow$ Linkage isomerism is due to presence of $SCN^{\ominus}$

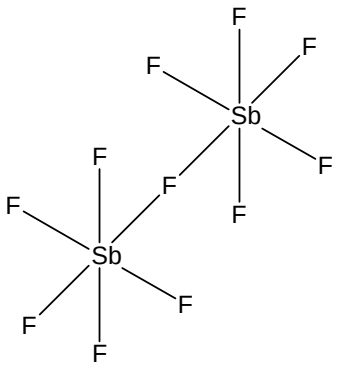
$\Rightarrow$ Optical isomerism is exhibited due to presence of $[Co(ox)_3]^{3-}$ which has asymmetric structure.

$\Rightarrow$ Geometrical isomerism is exhibited by $[Co(SCN)_2(NH_3)_4]^+$ part

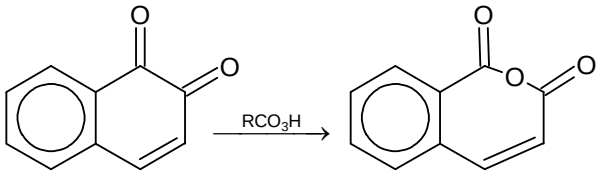
SECTION – C

1. $$AgNO_3 + H_2S_2O_3 \longrightarrow \underset{\text{(Black ppt)}}{Ag_2S}$$

2.



Sb-hybridisation is sp^3d^2 = total 6 hybrid orbitals per sb-atom.

3. 

Product (X)

The number of sp^2 – carbon is X are 10, the sum 1 + 0 = 1

4. $nM_{(s)} + nX^- \longrightarrow M_nX_n + ne^-$ (L.H.S)

For, $M_nX_n + ne^- \longrightarrow nM + nX^-$ (R.H.S)

$\therefore \quad E = \frac{-0.059}{n}\log_{10}\frac{0.01}{0.1}$

$\therefore$ n = 2

5. $\frac{1}{2}A_2 + \frac{1}{2}B_2 \longrightarrow AB$

$\Delta H = \frac{1}{2}e_{A-A} + \frac{1}{2}e_{B-B} - e_{A-B}$

$-100 = \frac{x}{2} + \frac{0.5x}{2} - x$

$\therefore$ x = 400 KJmol^{-1}

x = 4×10^2 KJmol^{-1}

Mathematics

PART – III

1. $\vec{u} = x\hat{i} + \sqrt{3}x\hat{j}$, $|u| = 2x$, $x > 0$

Now $|\bar{u}||\bar{u} - 2\hat{i}|^2 = |u - i|^2$

$2|x|\sqrt{(x-2)^2 + 3x^2} = \left((x-1)^2 + 3x^2\right)$

$\Rightarrow 4|x|\sqrt{x^2 - x + 1} = 4x^2 - 2x + 1$

Squaring, $16x^2(x^2 - x + 1) = 16x^4 + 4x^2 + 1 - 16x^3 - 4x + 8x^2$

$\Rightarrow 16x^2 = 12x^2 - 4x + 1$

$x = \dfrac{-4 \pm \sqrt{32}}{8} = \dfrac{-1 \pm \sqrt{2}}{2}$

$\Rightarrow x = \sqrt{2} - 1$

$\Rightarrow |\bar{u}| = \sqrt{2} - 1 = \sqrt{2} - \sqrt{1}$

$\Rightarrow a = 2, b = 1$

2. $2000x^6 + 100x^5 + 10x^3 + x - 2 = 0$

$\Rightarrow 2000x^6 + \dfrac{x\left((10x^2)^3 - 1\right)}{10x^2 - 1} - 2 = 0$

$\Rightarrow \dfrac{x(1000x^6 - 1)}{10x^2 - 1} = -2(1000x^6 - 1)$

$\therefore 1000x^6 - 1 = 0$ or $\dfrac{x}{10x^2 - 1} = -2$

$\Rightarrow x = -(10x^2 - 1)$

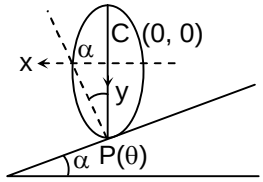
$\Rightarrow x^2 = \dfrac{1}{10}$ which is not possible

$\Rightarrow 20x^2 + x - 2 = 0$

$\Rightarrow x = \dfrac{-1 \pm \sqrt{161}}{40}$, $m = -1$, $n = 161$, $r = 40$

$\therefore (m + n + r) = 200$

3. Let $P(a\cos\theta, b\sin\theta)$



Slope of $cp = \dfrac{b}{a}\tan\theta$

$\tan\alpha = \dfrac{a^2 - b^2}{2ab}\sin 2\theta$

$\Rightarrow \sin 2\theta = \dfrac{2ab}{a^2 - b^2}\tan\alpha$

Using $\sin 2\theta \le 1$ and $e^2 = 1 - \dfrac{b^2}{a^2}$

$\dfrac{2\frac{b}{a}}{1 - \frac{b^2}{a^2}}\tan\alpha \le 1 \Rightarrow \dfrac{2\sqrt{1-e^2}}{e^2}\tan\alpha \le 1$

$\Rightarrow e^4 + 4e^2\tan^2\alpha - 4\tan^3\alpha \ge 0$

$e^2 \geq 2\tan\alpha(\sec\alpha - \tan\alpha) = \dfrac{2\sin\alpha}{1+\sin\alpha}$

4. $T_k = \cos^{-1}\left(\dfrac{1}{k(k+1)} + \dfrac{\sqrt{(k-1)k(k+1)(k+2)}}{k(k+1)}\right)$

Let $x = \dfrac{1}{k}$, $y = \dfrac{1}{k+1}$

$\sqrt{1-y^2} = \sqrt{1-\dfrac{1}{(k+1)^2}} = \dfrac{\sqrt{(k+1)^2-1}}{k+1} = \dfrac{\sqrt{k(k+2)}}{k+1}$

$T_k = \cos^{-1}\left(\dfrac{1}{k+1}\right) - \cos^{-1}\dfrac{1}{k}$, substituting n = 2, 3, 4

$S = \lim\limits_{n\to\infty} \cos^{-1}\left(\dfrac{1}{n+1}\right) - \cos^{-1}\dfrac{1}{2} = \dfrac{\pi}{2} - \dfrac{\pi}{3} = \dfrac{\pi}{6}$

$S_n = \dfrac{\pi}{6} = \dfrac{120\pi}{k}$

$\Rightarrow$ k = 720

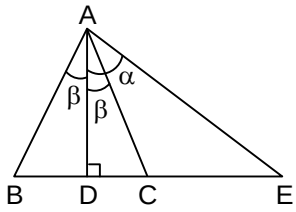
5. $\angle EAC = \alpha - \beta$

$\angle EAB = \alpha + \beta$

It is given that $\tan(\alpha - \beta)$, $\tan\alpha$ and $\tan(\alpha + \beta)$ form a G.P.

Thus $\tan^2\alpha = \tan(\alpha-\beta)\tan(\alpha+\beta) = \dfrac{\tan^2\alpha - \tan^2\beta}{1-\tan^2\alpha\tan^2\beta}$

$\tan\alpha = 1$, $\alpha = 45^\circ$, thus



AD = DE = $5\sqrt{2}$, so that area Δ(ABC) = $\dfrac{1}{2}BC \cdot AD = CD \cdot AD = 50\tan\beta$

Now, $\cot\alpha$, $\cot(\alpha - \beta)$, $\cot\beta$ form an A.P.

$2\cot(45^\circ - \beta) = 1 + \cot\beta \Rightarrow \cot\beta = 3$

$\Rightarrow \Delta(ABC) = \dfrac{50}{3}$

6. ABCD is a trapezium and its area = $\dfrac{1}{2}(a+b)h$

Where $\dfrac{a+b}{2} = EF$ (mid parallel)

Area = 2r(EF)

Now equation of EF is y = –x + c ..... (1)

From equation (1) passes through (r, r)

$\Rightarrow$ c = 2r

x + y = 2r, hence E = (2r, 0) and F(0, 2r)

$\therefore$ $EF = \sqrt{4r^2 + 4r^2} = 2\sqrt{2}r$

$\therefore$ Area ABCD = $2r \cdot 2\sqrt{2}r = 4\sqrt{2}r^2$

$\therefore$ $4\sqrt{2}\ r^2 = 900\sqrt{2}$, $r^2 = 225 \Rightarrow r = 15$

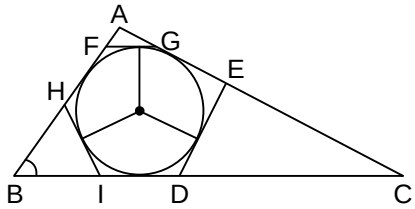
7. $A^2 - 2A + 2I = 0$

Divide B by $A^2 - 2A + 21$

$\Rightarrow B = (A^2 - 2A + 21)(f(A)) + A - I$

$\Rightarrow B = A - I = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$

8. FG = p
DE = q
HI = r



In ΔAFG $\frac{P}{\sin A} = \frac{AG}{\sin B} = \frac{AF}{\sin C}$

$\Rightarrow AG = \frac{bp}{a}$, $AF = \frac{cp}{a}$

Now, incircle of ΔABC is the ex–circle opposite to A for ΔAFG the semi perimeter of ΔAFG is

$S_{\Delta AFG} = \frac{1}{2}\left(p + \frac{bp}{a} + \frac{cp}{a}\right) = \frac{Sp}{a}$

$\Rightarrow r = S_{\Delta AFG} \tan\frac{A}{2} = \frac{Sp}{a}\tan\frac{A}{2}$

Since $r = (s - a)\tan\frac{A}{2}$

$\Rightarrow s - a = \frac{sp}{a} \Rightarrow \frac{p}{a} = 1 - \frac{a}{s}$

Similarly $\frac{q}{b} = 1 - \frac{b}{s} \Rightarrow \frac{r}{c} = 1 - \frac{c}{s}$

9. Since the coefficient of $\hat{k}$ in both $\vec{A}$ and $\vec{B}$ are the same, the only way that $\vec{A}$ and $\vec{B}$ can be parallel is that $f(t) = -f''(t)$, $f'(t) = g(t)$
The first differential equation, $f''(t) = -f(t)$

$\Rightarrow f(t) = p\cos t + q\sin t.\ (p, q \in R) = \sqrt{p^2 + q^2}\sin(t + \phi)$, $\tan\phi = \frac{p}{q}$

$f_{max}(t) = M = \sqrt{p^2 + q^2}$

$|\vec{A}| = \sqrt{f^2(t) + g^2(t) + 1} = \sqrt{p^2 + q^2 + 1}$

$|\vec{A}|$ is constant

10. A upon simplification $(a + b + c)(a - b)(b - c)(c - a)$
The only way this can be 0 is $(a + b + c) = 0$
$\Rightarrow$ fixed point is (1, 1)

11. $\int_0^x t\, f(x - t)\,dt = \int_0^x f(t)\,dt + \sin x + \cos x - x - 1$

$\Rightarrow \int_0^x (x - t) f(t)\,dt = \int_0^x f(t)\,dt + \sin x + \cos x - x - 1$

$\Rightarrow \int_0^x f(t)\,dt - \int_0^x t\, f(t)\,dt = \int_0^x f(t)\,dt + \sin x + \cos x - x - 1$

Again differentiating
$x f(x) + f(x) + f(x) - x f'(x) - f(x) > f'(x) - \sin x - \cos x$
$f(x) = f'(x) - \sin x - \cos x$
Solving this linear differential equation
We get, $f(x)\, e^{-x} = e^{-x}\cos x + c$
If $x = 0$, $f(0) = 0 \Rightarrow f(x) = e^x - \cos x$

12. Let w be the event that the ball drawn is white

Then $P(w) = \frac{x}{y}$

When, ball is lost $P(w) = \frac{x}{y}\frac{x-1}{y-1} + \frac{y-x}{y}\frac{x}{y-1} = \frac{x}{y}$

When 2 balls is lost $P(w) = \frac{{}^{x}C_2}{{}^{y}C_2}\frac{x-2}{y-2} + \frac{{}^{x}C_1\ {}^{y-x}C_1}{{}^{y}C_2}\frac{x-1}{y-2} + \frac{y-\ {}^{x}C_2}{{}^{y}C_2}\frac{x}{y-2} = \frac{x}{y}$

Similarly for n = 3, $P(w) = \frac{x}{y}$

13. $|w| = \frac{|v||u-z|}{|\bar{u}z-1|} = \frac{|u-z|}{|\bar{u}z-1|}$

Let, $|w| \le 1$

$\Rightarrow |u - z| \le |\bar{u}z - 1|$

$\Rightarrow (u - z)(\bar{u} - \bar{z}) \le (u\bar{z} - 1)(\bar{u}z - 1)$

$\Rightarrow (|u|^2 - 1)(|z|^2 - 1) \ge 1$

$\Rightarrow |z|^2 - 1 \le 0$

14. $f(0^-) = 0$, $f(0^+) = 1$, $f(0) = 1$

$\Rightarrow$ discontinuous at x = 0

$f(1^-) = 1$, $f(1) = 1$, $f(1^+) = 0$

$\Rightarrow$ discontinuous at x = 1

15. Squaring and adding

$\Rightarrow \sin(x + y) = 1$

$\Rightarrow x + y = (4n+1)\frac{\pi}{2}$, $n \in I$

$\Rightarrow x + y = \frac{\pi}{2}$, $y = \frac{\pi}{2} - x$

$\Rightarrow 5 \sin x = 3$

$\Rightarrow \cos x = \frac{4}{5}$

Also, $\cos y = \frac{3}{5}$ and $\sin y = \frac{4}{5}$

Hence, y > x

SECTION – C

1. $f(x) = \frac{1}{\sqrt{n^2}} + \frac{1}{\sqrt{n^2+2}} + ... + \frac{1}{\sqrt{n^2+2n+1}}$

Terms of the sequence are decreasing and number of terms are (2n + 2)

$\frac{2n+2}{\sqrt{n^2+2n+1}} \le f(x) \le \frac{2n+2}{\sqrt{n^2}}$

Now $\lim_{n\to\infty} \frac{2(n+1)}{\sqrt{n^2+2n+1}} = \lim_{n\to\infty} \frac{2n\left(1+\frac{1}{n}\right)}{n\sqrt{1+\frac{2}{n}+\frac{1}{n^2}}}$

Similarly $\lim_{n\to\infty}\frac{2(n+1)}{\sqrt{n^2}} = \lim_{n\to\infty}\frac{2(n+1)}{n} = 2$

2. $I_1 = \int_0^{\pi/2}(\sin x + r\cos x)^3\,dx = \int_0^{\pi/2}\sin^3 x + r^3\cos^3 x + 3r\sin^2 x\cos x + 3r^2\sin x\cos^2 x\,dx$

$= \frac{2}{3} + \frac{2r^3}{3} + r^2 + r$

$I_2 = \int_0^{\pi/2} x\cos x\,dx = (x\sin x + \cos x)_0^{\pi/2} = \frac{\pi-2}{2}$

$I = I_1 + \frac{4r}{\pi-2}I_2 = 2$

$\Rightarrow 2r^3 + 3r^2 - 3r + 2 = 6$

$\Rightarrow 2r^3 + 3r^2 - 3r - 4 = 0$

$\sum r_1 = -\frac{3}{2},\ \sum r_1 r_2 = -\frac{3}{2},\ \sum r_1^2 = \frac{21}{4}$

3. $A \rightarrow$ event that die A was used in first two throws
$B \rightarrow$ event that die B was used in first two throws
$C \rightarrow$ event that next throws gives red face
$E \rightarrow$ event that first two throws give red faces

$$P\left(\frac{A}{E}\right) = \frac{\frac{1}{2}\times\left(\frac{4}{6}\right)^2}{\frac{1}{2}\times\left(\frac{4}{6}\right)^2 + \frac{1}{2}\times\left(\frac{2}{6}\right)^2} = \frac{4}{5}$$

$P\left(\frac{B}{E}\right) = \frac{1}{5} \Rightarrow P\left(\frac{C}{E}\right) = \frac{3}{5}$

a = 3, b = 5

4. $f(x) = 2f\left(\frac{x^2}{2}\right) + f(6-x^2)$

$\Rightarrow f'(x) = 2f'\left(\frac{x^2}{2}\right)x - 2xf'(6-x^2)$

$\Rightarrow f'(x) = 2x\left(f'\left(\frac{x^2}{2}\right) - f'(6-x^2)\right)$

$f'\left(\frac{x^2}{2}\right) > f'(6-x)^2$ (if $\frac{x^2}{2} > 6-x^2$)

$\because$ f′(x) is increasing

$\frac{x^2}{2} > 6-x^2 \Rightarrow x^2 > 4$

$\Rightarrow f'\left(\frac{x^2}{2}\right) - f'(6-x^2) > 0$ when x < –2 or x > 2

$\Rightarrow$ f′(x) > 0, when $x \in (-2, 0) \cup (2, \infty)$

$\Rightarrow a + b + c = 0$

5. $R = 1 + \sin\theta,\ \theta = \frac{\pi}{8}$

$\Rightarrow c_1c_2 = R + r$

$\therefore\ \sin\theta = \frac{R-r}{R+r}$

$R \sin\theta + r \sin\theta = R - r$

$R(1 - \sin\theta) = r(1 + \sin\theta)$ (Put $R = 1 + \sin\theta$)

$\Rightarrow (1 + \sin\theta)(1 - \sin\theta) = r(1 + \sin\theta)$

$\Rightarrow r = 1 - \sin\theta$

$\Rightarrow Rr = 1 - \sin^2\theta = \cos^2\theta$

$\Rightarrow \cos 2\theta = \frac{1}{2}(1+\cos\theta) = \frac{1}{2}\left(1+\cos\frac{\pi}{4}\right) = \frac{2+\sqrt{2}}{4}$

$\Rightarrow a = 2, b = 4$